

Weekly Self-Study Tasks to Complete

Phase 2 — Week 1: JavaScript Fundamentals for Leaders

This document covers all of Week 1 of Phase 2 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote Page lab, where you will make the page calculate a live premium estimate, validate input, and load recent quotes asynchronously. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~58 min)

Phase 1 introduced what JavaScript is and where it runs at a concept level. This week you build on that by watching real code run. Watch the specific resources called out below — not entire courses — and keep the Chrome DevTools Console open as you go: each one reinforces the functions, DOM updates, events, and async calls you saw in the instructor session. Every link and duration below has been verified against the source platform.

Topic	Material Description	Est. Time	Link / Reference
Functions & Objects	JavaScript: Getting Started (Mark Zamoyta, Pluralsight, updated Jun 28, 2025) — watch the Functions module. Shows real functions being written and called: parameters, return values, and reuse — the same <code>calculatePremium()</code> idea from the session. If you have time, the Objects and the DOM module is a natural next watch. Code-focused and beginner-friendly.	~23 min	pluralsight.com — JavaScript: Getting Started

Topic	Material Description	Est. Time	Link / Reference
The DOM & Events	Dynamic scripting with JavaScript (MDN Web Docs, updated Oct 2025) — read and run the “DOM scripting introduction” and “Introduction to events” articles. Reinforces querySelector, changing textContent, and wiring up input and submit handlers — exactly what powers the live estimate. Free and authoritative.	~15 min	developer.mozilla.org — Dynamic scripting
Async, Promises & Fetch	Asynchronous JavaScript (MDN Web Docs, updated Jul 2025) — read and run “Introducing asynchronous JavaScript” and “How to use promises” (which introduces async/await). Explains why a slow network call should not freeze the screen and how fetch loads data — the recent-quotes feature in the lab. Free and authoritative.	~15 min	developer.mozilla.org — Asynchronous JavaScript
Slide Skim	Review the week's slide deck at a high level. Confirm you recognize function, object, arrow function, template literal, the DOM, event listener, preventDefault, try/catch, async/await, and fetch. No need to memorize syntax — this is a check that the session and the videos together left you fluent in the vocabulary you'll see in any JavaScript file. Skim only.	~5 min	See course material

Practice Activities (1 hour)

Complete this week's hands-on lab: "Evergreen Insurance Quote Page — Live Premium Estimate with JavaScript." The starter is the same Evergreen page you built up in Phase 1. This week you will turn that static page into an interactive one: a premium that updates as the user types, input validation that fails gracefully, and a recent-quotes list loaded from a data file with `async/await`. The whole exercise takes about one hour.

In the lab you will:

- Open the `Labfiles/evergreen-quote-live-estimate` folder in VS Code and run it with Live Server.
- Write a `calculatePremium()` function that returns a monthly premium from coverage type, age, and amount, using an object as the rate lookup table.
- Write a `validateInputs()` function that throws a clear Error for bad input, such as an out-of-range age.
- Fill in `updateEstimate()` with `try/catch` so the page shows a friendly message instead of crashing.
- Wire up the form's input and submit events, using `e.preventDefault()` to stop the page reload.
- Watch the estimate update live as you type, and watch validation reject an invalid age.
- Load `quotes.json` with `async/await` and `fetch`, then render the recent-quotes list.
- Break the `fetch` on purpose (rename the data file) and observe the contained, friendly failure.
- Review your changes with `git diff`, then commit with a clear message such as "Add live premium estimate, validation, and recent quotes."

The lab instructions (with "Why this matters" callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for leaders who want to go deeper beyond the core curriculum. They sit outside the 4-hour weekly budget. None of these are required — you can absolutely stop at the resources above and the lab and have everything you need for the rest of Phase 2. TypeScript and React build directly on this week and are covered next, so focus for now on functions, the DOM, events, and `async`.

Topic	Material Description	Link / Reference
JavaScript (Full Course)	JavaScript: Getting Started — Mark Zamoyta (Pluralsight, updated Jun 28, 2025). 3h 58m, Beginner. The complete course behind the Functions module above: variables and constants, types and operators, program flow, functions, objects and the DOM, arrays, and scope. The full-length path if you want one guided tour of the language (Video Course)	pluralsight.com — JavaScript: Getting Started
JavaScript (Video)	JavaScript Essential Training — Morten Rand-Hendriksen (LinkedIn Learning). 6h 14m, Beginner. Covers objects, the DOM, arrays, functions, events, and troubleshooting, with interactive CoderPad code challenges. A thorough second pass in a different voice (Video Course)	linkedin.com/learning — JavaScript Essential Training
JavaScript Reference	MDN JavaScript Guide — developer.mozilla.org. The canonical free reference for JavaScript syntax and semantics, with runnable examples and deep-links to every keyword and built-in. Keep it open when you are reading a PR with unfamiliar syntax (Reference)	developer.mozilla.org — JavaScript Guide
Async & Fetch	Using the Fetch API — MDN Web Docs (updated Aug 2025). Deeper coverage of network requests: checking response status, reading JSON, and handling errors — the production reality behind the lab's recent-quotes call (Reference)	developer.mozilla.org — Using the Fetch API
Free Book	Eloquent JavaScript, 4th edition — Marijn Haverbeke (2024). Free to read online under a Creative Commons license. The	eloquentjavascript.net

Topic	Material Description	Link / Reference
	chapters on functions, data structures, the DOM, handling events, and asynchronous programming map directly to this week. The rest goes deeper and is optional (Book)	
Interactive	freeCodeCamp — JavaScript Algorithms and Data Structures (current curriculum). Free, browser-based exercises you complete by typing code: variables, functions, objects, arrays, and loops. Useful if you learn by doing rather than watching (Interactive)	freecodecamp.org — JavaScript certification
Browser Support	Can I Use — up-to-the-day browser-support matrix for every JavaScript and web feature. Useful when an engineer tells you a feature “isn't safe to ship yet” — you can check the matrix for yourself (Tool)	caniuse.com